evolution, and so on. It is not a scientific demonstration. The question is among the most fascinating in all of science. As described in this book, we are just developing the tools to treat it seriously.

What about the related matter of whether we are capable of creating intelligences smarter than ourselves? Computers routinely do mathematics that no unaided human can manage, outperform world champions in checkers and grand masters in chess, speak and understand English and other languages, write presentable short stories and musical compositions, learn from their mistakes, and competently pilot ships, airplanes, and spacecraft. Their abilities steadily improve. They're getting smaller, faster, and cheaper. Each year, the tide of scientific advance laps a little further ashore on the island of human intellectual uniqueness with its embattled castaways. If, at so early a stage in our technological evolution, we have been able to go so far in creating intelligence out of silicon and metal, what will be possible in the following decades and centuries? What happens when smart machines are able to manufacture smarter machines?

PERHAPS THE CLEAREST INDICATION that the search for an unmerited privileged position for humans will never be wholly abandoned is what in physics and astronomy is called the Anthropic Principle. It would be better named the Anthropocentric Principle. It comes in various forms. The "Weak" Anthropic Principle merely notes that if the laws of Nature and the physical constants—such as the speed of light, the electrical charge of the electron, the Newtonian gravitational constant, or Planck's quantum mechanical constant—had been different, the course of events leading to the origin of humans would never have transpired. Under other laws and constants, atoms would not hold together, stars would evolve too quickly to leave sufficient time for life to evolve on nearby planets, the chemical elements of which life is made would never have been generated, and so on. Different laws, no humans.